

Hang Yu

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Overview

My current research focuses on **Human-Centered AI** in **Robotics**, specifically, on learning from multi-modular teachings, including human feedback (**RLHF**), demonstrations (**LfD**), and generative methods(**VLM**, **VLA**). Before that, I worked on Recommendation Systems for three years and competitive programming for seven years.

Skills

Robotic Skills: ROS, ROS2, Gazebo, Isaac Sim, Gello, UMI, VLA, Sim2Real

Robotic Hardware: Kinova Gen2/Gen3/Gen3-lite, ABB IRB 1300, Fetch, UR3E, UR5, Misty II, Cozmo

Machine Learning Skills: RL, Inverse RL, RLHF, LfD, RecSys, Imitation Learning, LLM/VLM/VLA Training

Education

Tufts University

Ph.D. in Computer Science, GPA: 4.0/4.0

Advisor: Dr. Elaine Schaertl Short

Medford, MA

August 2026(expected)

Tufts University

M.S.E. in Computer Science, GPA: 3.95/4.0

Advisor: Dr. Elaine Schaertl Short

Medford, MA

Jan. 2021

Research/Work Experience

Research Intern, **ABB Robotics**

June 2026 – Present

Conduct research on VLAs for bimanipulation.

Improve VLA for long horizon manipulation tasks using task progress estimator and embodied memory.

Research Scientist Intern, **Berkshire Grey, SoftBank Group**

Jan 2026 – June 2026

Work on robot learning and developing VLA for bimanipulation robotic arms.

Build Gello and UMI pipeline for collecting data for VLA training and fine-tuning.

Research Assistant, Assistive Agent Behavior and Learning Lab, Tufts University

2020 – present

Conduct research on robot learning and human-in-the-loop learning.

Robot learning from non-expert users with data collected in real-world scenarios in public spaces.

Publications

Peer-Reviewed Publications

- C6 **Hang Yu**, James Staley, Shijie Fang, Jindan Huang, Wenchang Gao, Reuben M. Aronson, and E. Short, *PHIRL: Aligning Rewards with Progress for Inverse Reinforcement Learning with Vision-Language-Model-in-the-Loop*, (**CoRL**), 2026 (under review).
- DC1 **Hang Yu**, Elaine Short, *Enabling Robust Learning from Non-Experts by Leveraging Human Demonstrations and Human Feedback*. IEEE International Conference on Robotics Automation **ICRA**, Doctoral Consortium, 2025, Atlanta, USA
- C5 **Hang Yu**, Reuben M. Aronson, Katherine H. Allen, and E. Short, *From “Thumbs Up” to “10 out of 10”: Reconsidering Scalar Feedback in Interactive Reinforcement Learning* 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), Detroit, USA, 2023.
- C4 **Hang Yu**, Qidi Fang, Shijie Fang, Reuben M. Aronson, and E. Short, *How Much Progress Did I Make? An Unexplored Human Feedback Signal for Teaching Robots* 2024 IEEE **RO-MAN**, Pasadena, USA
- C3 Matthew Ebisu*, **Hang Yu***, Reuben M. Aronson, Elaine S. Short, *See What I Mean? Expressiveness and Clarity in Robot Display Design*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.
- C2 Qidi Fang*, **Hang Yu***, Shijie Fang, Jindan Huang, Qiuyu Chen, Reuben M. Aronson, Elaine S. Short, *CHARM: Considering Human Attributes for Reinforcement Modeling*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.
- C1 Shijie Fang*, **Hang Yu***, Qidi Fang, Reuben M. Aronson, Elaine Schaertl Short, *Demonstration Sidetracks: Categorizing Systematic Non-Optimality in Human Demonstrations*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.

(*) means these authors contributed equally to the work.

Journal Publications.....

J3 Tan, Z., **Yu, H.**, Wei, W., & Liu, J.. *Top-K interesting preference rules mining based on MaxClique*. **Expert Systems with Applications**, 143, 113043.

J2 **YU Hang**, WEI Wei, TAN Zheng, LIU Jing-lei. *Contextual Preference Collaborative Measure Framework Based on Belief System*. **Computer Science**, 2020, 47(4): 74-84.

J1 TAN, Z., LIU, J., & **YU, H.**. *Conditional preference mining based on MaxClique*. **Computer Applications**, 37(11), 3107.

Selected Workshop Papers.....

W2 **Hang Yu**, James Staley, Shijie Fang, Wenchang Gao, Reuben M. Aronson, and Elaine S. Short *PHIRL: Progress Heuristic for Inverse Reinforcement Learning*. RSS workshop 2025: Continual Robot Learning from Humans

W1 **Hang Yu** and Elaine Schaertl Short. 2021. *Active Feedback Learning with Rich Feedback*. HRI '21 Companion. Association for Computing Machinery, New York, NY, USA, 430–433.

Projects.....

- A World Model Powered VLA with Hierarchical Self-Attention for Manipulation**
Integrating Hierarchical Self-Attention into GigaBrain-0 to achieve better grasping and object manipulation.
Github: https://github.com/HangYu8123/gigabrain_with_Hierarchical_Self_Attention
- Harness Coding Workflow, GitHub 400+ Star★**
Harness for agentic coding workflow for coding, debugging, refactoring, and logging. (GitHub 200+ Stars)
Github: <https://github.com/HangYu8123/HarnessFlow.git>
- PHIRL: Progress-Heuristicized Inverse Reinforcement Learning**
An IRL learning framework that learns from human feedback, human demonstrations, and VLMs.
Github: <https://github.com/HangYu8123/PHIRL-Progress-Heuristic-for-Inverse-Reinforcement-Learning>

Honors/Awards.....

ICRA Doctoral Consortium	2025
Selected to participate in the ICRA Doctoral Consortium.	
RO-MAN Student Grant	2024
Selected as the winner of the travel grant of RO-MAN 2024.	
Yantai University Outstanding Student Scholarship	2017
Awarded to students at Yantai University who had exceptional performance in academia and competitions.	
Lanqiao Programming Competition National First Prize	2016
Awarded to participants who were Top 0.5% among all the participants.	

Selected Professional Service.....

Mentor	
Isabella Bock, Undergraduate Student, Summer Intern, now at Tufts University	
Matthew Ebisu, Master’s Student, Co-author of C3, now at MassRobotics	
Shijie Fang, Master’s Student, Co-author of C4, C1, C2, now at Dongnan University	
Teaching Assistant	
Ethics for AI, Robotics, and Human-Robot Interaction, Tufts University	Spring 2024, Spring 2025
Human-Robot Interaction, Tufts University	Fall 2022
Selected Service	
ICRA Undergraduate Research Panelist	2025
Keynote Presenter at First Robotics Competition Conference NE	2024
DIAMOND Program Panelist	2023
Tufts Computer Science Student Council	2023 – Present
Program Committee for AAMAS	2026
Reviewer for International Conference on Human-Robot Interaction (HRI)	2021, 2022, 2023, 2024
Reviewer for International Conference on Robotics and Automation (ICRA)	2023, 2024, 2025